9 Polynomial Operations

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RULE · Add / subtract polynomials (ko'phad)

Combine like terms. Subtracting: flip every sign in the second polynomial – not just the first.

$$(5x^3 - 2x + 1) - (2x^3 + 3x - 4) = 3x^3 - 5x + 5$$

$x^2 + x^2 = 2x^2$ – exponents don't change when adding, only when multiplying.

RULE · Multiplying polynomials

Every term in the first factor times every term in the second, then combine.

$$(x + 4)(2x^2 - x + 3) = 2x^3 + 7x^2 - x + 12$$

RULE · Dividing polynomials

By a monomial: divide each term separately. By a binomial: long division, like numbers.

$$(x^2 + 5x + 6) \div (x + 2) = x + 3 \quad \text{(remainder 0)}$$